Kea Francis

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Summary Statement

Engineer specializing in robotics with 5+ years of hands-on experience diagnosing, debugging, and improving complex robotic systems across 6-DOF industrial robotics and autonomous mobile platforms. Currently pursuing an M.S. in Robotics Engineering at WPI, with active research in humanoid robotics and perception-based AI. Experienced working on electromechanical system integration with a strong foundation in ROS2, Python, C++, and Linux-based environments.

Technical Skills

Programming Languages: Python, C/C++, SQL, Bash, VBA, and MATLAB

Tools: Robotic Operating System (ROS2), JIRA, Raspberry Pi, Arduino, Particle, Gazebo/RViz, OpenCV

Operating Systems: Linux (Raspbian and Ubuntu), Mac and Windows

Software: Rhinoceros CAD, Microsoft Access, SolidWorks

Debug & Evaluation: Confusion matrix evaluation, root cause analysis

Robotics Projects & Research

- **Automated Diagnostic Pipeline** (Amazon Robotics, May 2024 – March 2025) Developed to address gaps in ML tool output that translated operational robotic failures into actionable datasets for AI research teams, accelerating model retraining cycles.

- **Resource Link Automation Tool** (Amazon Robotics, January 2026 – March 2026) Designed a resource link automation tool that adds relevant resources to trouble tickets reducing time to resolution by 20 minutes per ticket.

- **Misroute Classification System** (Amazon Robotics, January 2025 – Present) Designed to improve ticket resolution accuracy across support workflows, built weighted keyword scoring and performance tracking modules, iteratively tuned for decision accuracy.

- **Master's Thesis - Adaptive Gamification for Dementia Care** (WPI RoboCare Lab, August 2024 – Present) Designing modular engagement modules for Pepper humanoid robot using Python, JavaScript, React, and dockerized ROS2, with MediaPipe-based perception for real-time body tracking and adaptive interaction to support physical activity in dementia care. Integrated LLM-based response generation via ChatGPT API to enable context-aware adaptive dialogue.

- **Autonomous Cart Diagnostic Tool** (Amazon Robotics AI via Experis, May 2021 – April 2022) Built Python tool to SSH into autonomous carts and retrieve diagnostics, cutting access time from 2 minutes to 10 seconds

- **Rapiro Social Robot Interface** (UTK Research, September 2017 – May 2019) Built Bluetooth communication and data-sharing interface on Raspberry Pi 3 for pediatric speech therapy applications.

Robotics Experience

Technical Support Engineer at Amazon Robotics (May 2022 – Present)

- Diagnose and debug robotic system failures using Linux-based logging tools and cross-functional investigations to inform system-level improvements
- Mentor new hires and perform as a technical lead for L5-tier support engineer ⅗ days each week
- Contributed to CI/CD through troubleshooting issues caused by software deployments

- Served as the primary technical point of contact for internal and external stakeholders, leading root cause analysis and resolution of complex robotic system failures.

Technical Support Technician at Amazon Robotics AI via Experis (May 2021 – April 2022)
- Owned nightly software deployments and validation in a robotics CI/CD test environment.
- Debugged robotic software/hardware across local and remote Seattle fleet; logged issues via JIRA.
- Boosted test suite completion rate from 7 days to 1 day.
- Improved QA process through analysis, reporting, and documentation updates.

Junior Robotics Engineer at CleanRobotics (October 2020 – January 2021)
- Developed C++ programs on Particle for watchdog timers on electromechanical components, optimized motor and actuator acceleration through drive parameter tuning, and remote troubleshooting tools.

Robotics Technician at Dal-Tile Corporation (November 2019 – August 2020)
- Led bring-up and integration of newly deployed robotic arms (Rhinoceros CAD) in an active manufacturing environment, leading control validation and human-in-the-loop testing
- Developed custom Computerized Maintenance Management System (CMMS) GUI using SQL, VBA, and MS Access

Engineering Experience

Automation Engineer at Nexus Bioenergy (Part Time) (May 2021 – Dec 2021)
- Implemented an Arduino-based PLC system with C++ programming for data collection, analysis, and transfer to Excel.

Oak Ridge National Laboratory — Reliability & Maintainability Intern (May 2016 – May 2017)
- Led weekly FMEA sessions for the Summit supercomputer; built a Mechanical-Asset catalog and KPIs that improved maintenance-plan coverage for critical cooling and power systems.
- Optimized the FSC CMMS for asset tracking and predictive-maintenance scheduling, enhancing uptime on complex electromechanical equipment.

Education

Worcester Polytechnic Institute | Master of Science in Robotics Engineering **| January 2025 – Expected August 2026 | Relevant Courses:** Robot Dynamics, Robot Control, ML for Robotics, Human-Robot Interaction, Systems Engineering

Worcester Polytechnic Institute | Robotics Engineer Graduate Certificate **| May 2024 – May 2025**

The University of Tennessee Knoxville | Bachelor of Science in Mechanical Engineering | **July 2015 – May 2019**

Certifications

Associate Systems Engineering Professional (ASEP) | INCOSE | **Issued Nov 2024**

Intro to PostgreSQL Databases with PgAdmin For Beginners | Udemy | **Issued May 2021**

Level One Certified Airborne Ultrasound Inspector | Credential ID 2016-2592 | **Issued Sep 2016**

Presentations and Demonstrations

- **NHCGNE Conference Virtual Podium Presentation**, (2025, October). *Adaptive gamified robotic interventions for mitigating physical inactivity among persons with dementia.* Virtual podium presentation at the 2025 NHCGNE Annual Conference.
- **Live Demo: Adaptive Gamification in Dementia Care**, MassITC Conference. Demonstrated ROS2-based control and interruption protocols for social robotics